

- 7 (a) The radioactive isotope of Bismuth $^{210}_{83}\text{Bi}$ decays into Polonium (chemical symbol: Po) with the emission of a beta particle.

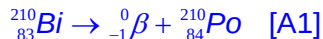
(i) State the origin of the beta particle.

Beta particle is emitted from the nucleus OR

Transmutation from one neutron in Bi : $^1_0n \rightarrow ^1_1H + ^0_{-1}\beta$ [B1]

[1]

(ii) Write down the equation representing the beta decay of $^{210}_{83}\text{Bi}$.



[1]

(iii) State two quantities that are conserved in any radioactive decay process.

1. total momentum/ total mass-energy/ total nucleon no/ total atomic no/ total charge/ proton no.

2. Don't accept no. of proton/electrons

[2]

Any two. [A2]

- (iv) The mass of a $^{210}_{83}\text{Bi}$ nucleus is $209.939u$. Show that its mass defect is $1.767 u$.
(mass of proton, $m_p = 1.00729 u$; mass of neutron, $m_n = 1.00867 u$)

$$\text{Mass of protons + neutrons} = [(83 \times 1.00729) + (127 \times 1.00867)]$$

$$= 211.706 u \text{ [M1]}$$

$$\text{Mass defect} = 211.706 u - 209.939 u \text{ [M1]}$$

$$\text{Mass defect} = 1.767 u \text{ [A0]}$$

[2]

- (v) Calculate the binding energy per nucleon, in MeV, of $^{210}_{83}\text{Bi}$.

$$\text{Binding energy} = 1.767 \times 1.66 \times 10^{-27} \times (3 \times 10^8)^2 = 2.64 \times 10^{-10} \text{ J [M1]}$$

$$\text{Binding energy per nucleon} = [2.64 \times 10^{-10} / 1.6 \times 10^{-19} \times 10^6] / 210 = 7.86 \text{ MeV [A1]}$$

binding energy per nucleon = MeV [2]

- (b) Fig. 7.1 shows the energy spectrum for beta particles emitted during the decay of $^{210}_{83}\text{Bi}$. The intensity indicates the number of beta particles emitted with each particular kinetic energy.

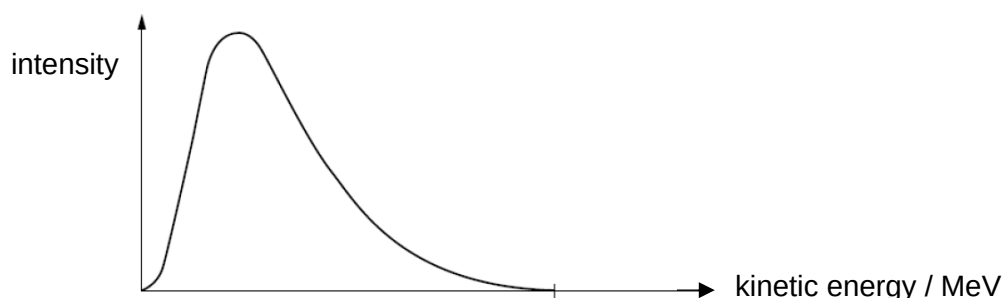


Fig. 7.1

Explain how a consideration of this kinetic energy spectrum provide evidence for the prediction of the existence of the antineutrino.

.....
There is a range of energies.

.....
Energy per decay is constant due to conservation of energy.[B1]

.....
Antineutrino has the remaining energy. [B1]

.....
.....
.....[2]

[Total: 10]

8 Read the passage below and answer the questions that follow